C3.2 How are metals with different reactivities extracted?	
Teaching and learning narrative	**Assessable learning outcomes** *Learners will be required to:*
Metals can be placed in an order of reactivity by looking at their reactions with water, dilute acid and compounds of other metals. The relative reactivity of metals enables us to make predictions about which metals react fastest or which metal will displace another. When metals react they form ionic compounds. The metal atoms lose one or more electrons to become positive ions. The more easily this happens the more reactive the metal. These reactions can be represented by word and symbol equations including state symbols. **Ionic equations show only the ions that change in the reaction and show the gain or loss of electrons. They are useful for representing displacement reactions because they show what happens to the metal ions during the reaction.** The way a metal is extracted depends on its reactivity. Some metals are extracted by reacting the metal compound in their ores with carbon. Carbon is a non-metal but can be placed in the reactivity series of the metals between aluminium and zinc. Metals below carbon in the reactivity series are extracted from their ores by displacement by carbon. The metal in the ore is reduced and carbon is oxidised. Highly reactive metals above carbon in the reactivity series are extracted by electrolysis. **Scientists are developing methods of extracting the more unreactive metals from their ores using bacteria or plants. These methods can extract metals from waste material, reduce the need to extract 'new' ores, reduce energy costs, and reduce the amount of toxic metals in landfill. However, these methods do not produce large quantities of metals quickly (IaS4).**	1. deduce an order of reactivity of metals based on experimental results including reactions with water, dilute acid and displacement reactions with other metals 2. explain how the reactivity of metals with water or dilute acids is related to the tendency of the metal to form its positive ion to include potassium, sodium, calcium, aluminium, magnesium, zinc, iron, lead, [hydrogen], copper, silver 3. use the names and symbols of common elements and compounds and the principle of conservation of mass to write formulae and balanced chemical equations **and ionic equations** 4. explain, using the position of carbon in the reactivity series, the principles of industrial processes used to extract metals, including the extraction of zinc 5. explain why electrolysis is used to extract some metals from their ores **6. evaluate alternative biological methods of metal extraction (bacterial and phytoextraction)**

Linked learning opportunities

Practical work:

- Investigate the reactivity of different metals with water and dilute acid
- Investigate the reactivity of Zn, Fe and Cu by heating each metal with oxides of each of the other two metals

Specification links:

- C1.1 introduces oxidation

Ideas about Science:

- impacts of metal extraction on the environment, the measures scientists are taking to mitigate them, and the risks, costs and benefits of different courses of action (IaS4)